Project Title

A Project Report

submitted in partial fulfillment of the requirements

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AICTE Internship on AI: Transformative Learning
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by

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We would like to take this opportunity to express our deep sense of gratitude to all individuals who helped us directly or indirectly during this thesis work.

Firstly, we would like to thank my supervisor, JAY RATHOD, for being a great mentor and the best adviser I could ever have. His advice, encouragement and the critics are a source of innovative ideas, inspiration and causes behind the successful completion of this project. The confidence shown in me by him was the biggest source of inspiration for me. It has been a privilege working with him for the last one year. He always helped me during my project and many other aspects related to the program. His talks and lessons not only help in project work and other activities of the program but also make me a good and responsible professional.

During this internship, I had an incredible learning experience, exploring various Python libraries and enhancing my technical skills. This journey was not just about building a healthcare assistant chatbot but also about strengthening my foundation in natural language processing (NLP) and AI.

I actively participated in weekly meetings, where I had the opportunity to clarify doubts, gain insights from industry experts, and improve my problem-solving approach. These interactions significantly enhanced my understanding of chatbot development, API integration, and machine learning models.

Working on this project helped me develop a structured workflow, from research and planning to implementation and testing. I explored libraries like **NLTK, Transformers, and Streamlit**, understanding how each contributes to chatbot functionality. Troubleshooting errors and optimizing responses further deepened my debugging skills.

Overall, this internship provided hands-on experience in AI-driven chatbot development, reinforced my confidence in Python programming, and motivated me to explore more advanced applications in healthcare technology.

ABSTRACT

Healthcare accessibility remains a challenge, especially for elderly individuals and those in remote areas who lack immediate medical consultation. Many people turn to the internet for health advice, often encountering unreliable information. To address this, we developed a Healthcare Assistant Chatbot, an AI-driven system designed to provide users with basic symptom analysis, medication guidance, and diet recommendations.

The objective of this project is to create a user-friendly chatbot that delivers instant, general health-related responses using Natural Language Processing (NLP). While it does not replace professional medical advice, it serves as a first-level assistant for common health inquiries.

The chatbot was developed using Python, Streamlit, and NLP models, including pre-trained AI models for text generation. It processes user queries, identifies key health-related terms, and generates relevant responses. The chatbot covers symptom-related concerns, general medication advice, and dietary recommendations based on predefined knowledge.

Key results show that the chatbot effectively provides informative and structured responses, helping users make more informed health decisions while emphasizing the importance of consulting a doctor for serious conditions.

In conclusion, this project enhances healthcare accessibility by providing quick, AI-powered health guidance. Future improvements include integrating real-time doctor consultations, expanding the knowledge base, and incorporating machine learning for more personalized responses.

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CHAPTER 1

Introduction

1.1 Problem Statement

Many people struggle to access immediate healthcare guidance, leading to misinformation and delayed treatment. This project aims to develop a Healthcare Assistant Chatbot to provide basic symptom analysis, medication advice, and diet recommendations, offering users quick and reliable health-related information.

1.2 Motivation

With the rise of AI in healthcare, this chatbot helps users get instant, general health advice before consulting a doctor. It benefits elderly individuals, people in remote areas, and those seeking quick guidance for minor health concerns, reducing misinformation and improving accessibility.

1.3 Objective

- Develop an AI-powered chatbot for basic health guidance.
- Provide user-friendly and informative responses using NLP.
- Ensure easy accessibility for all users.
- Encourage preventive care and timely medical consultations.

1.4 Scope of the Project

✅ Provides general health guidance on symptoms, medications, and diet.
❌ Does not diagnose illnesses or prescribe medications.
✅ Uses NLP for interactive responses.
❌ Not suitable for emergencies or critical conditions.

CHAPTER 2

Proposed Methodology

3.1 Requirement Specification

The tools and technologies required to implement the solution.

3.1.1 Hardware Requirements:

RAM: At least 4GB (8GB recommended for smooth execution)
GPU: If using more advanced models, a CUDA-compatible GPU can speed up text generation
OS: Windows, Linux, or macOS

3.1.2 Software Requirements:

Python >= 3.10 version

Streamlit – For building the chatbot UI

nltk – For natural language processing (tokenization, stopwords)

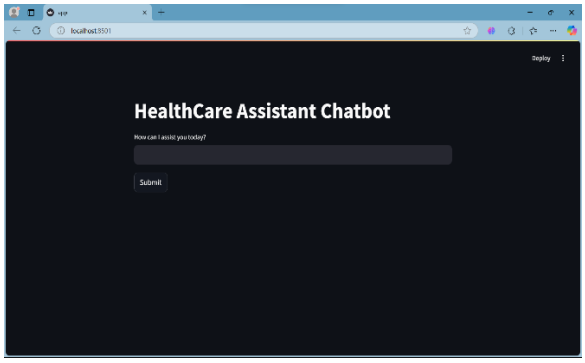
transformers – For text generation (using distilgpt2 model)

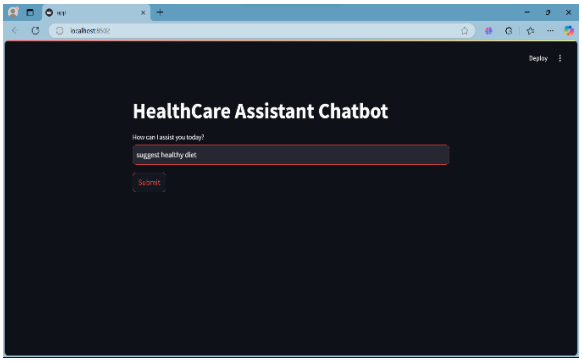
torch – Required for running models from transformers

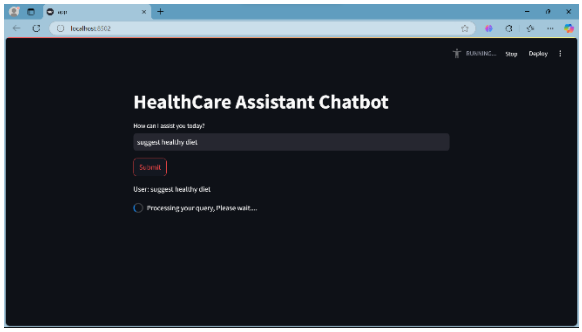
CHAPTER 3

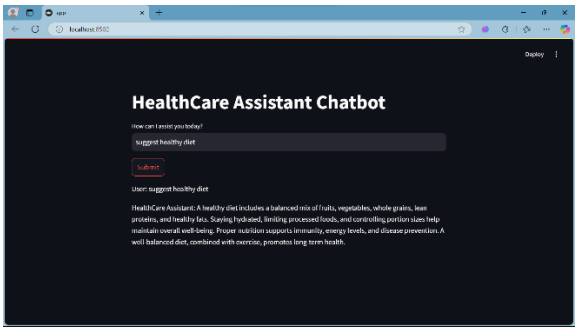
Implementation and Result

4.1 Snap Shots of Result:









4.2GitHub Link for Code:

https://github.com/RavishaShrivastava/Health_assistance_chatbot

OR

CHAPTER 4

Discussion and Conclusion

5.1 Future Work:

Medication & Diet Optimization – Use AI to suggest personalized medication schedules and diet plans based on health conditions.

AI-Powered Diagnosis & Recommendations – Implement machine learning models for more accurate symptom analysis and personalized health advice.

5.2 Conclusion:

The health assistant serves as a user-friendly tool for monitoring health conditions, providing basic symptom guidance, and offering general wellness recommendations. By integrating features like symptom checking, medication reminders, and diet suggestions, the assistant enhances accessibility to essential health information. While it does not replace professional medical advice, it helps users make informed decisions about their well-being and encourages proactive health management.

REFERENCES

[1]. Ming-Hsuan Yang, David J. Kriegman, Narendra Ahuja, “Detecting Faces in Images: A Survey”, IEEE Transactions on Pattern Analysis and Machine Intelligence, Volume. 24, No. 1, 2002.